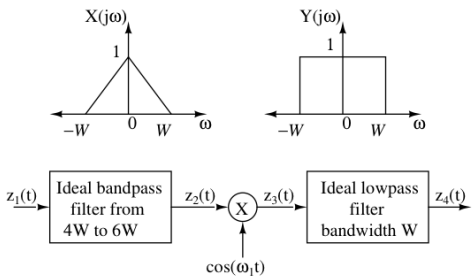
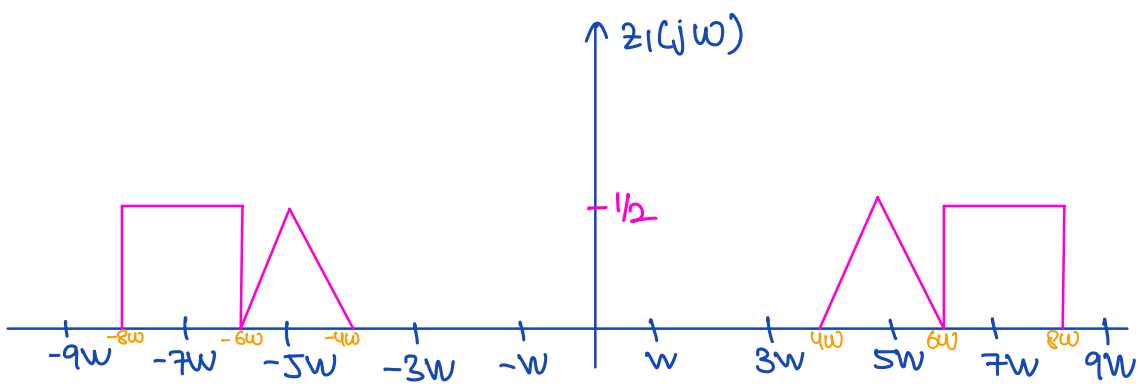
8. $x(t)$ and $y(t)$ have Fourier transforms as shown below. Sketch the Fourier transform of the various signals $z_i(t)$ for $i = 1, 2, 3, 4$ in the system shown below given that $z_1(t) = x(t)\cos(\omega_1 t) + y(t)\cos(\omega_2 t)$. Determine $z_4(t)$ in terms of $x(t)$ and $y(t)$? Assume that $\omega_1 = \omega_2 - 2W = 5W$. $\omega_1 = 5W \quad \omega_2 = 7W$

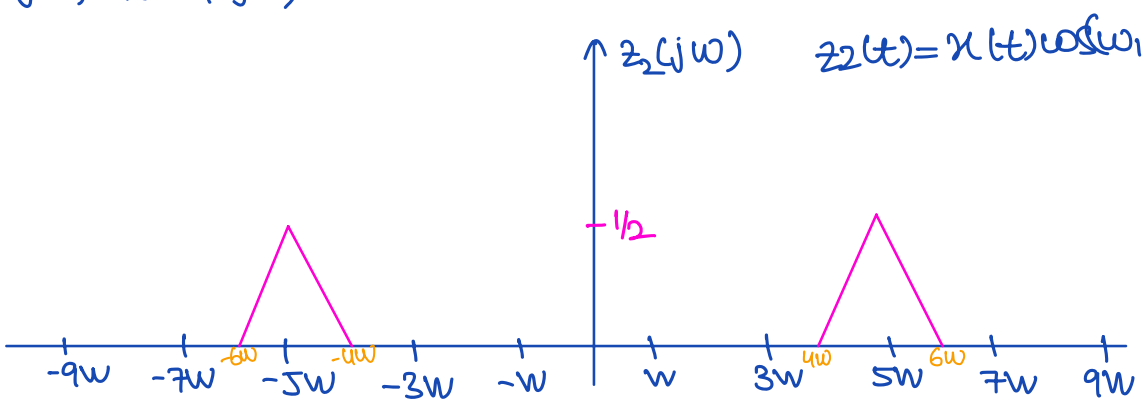
$$z_1(t) = x(t)\left[\frac{e^{j\cdot 5Wt} + e^{-j\cdot 5Wt}}{2}\right] + y(t)\left[\frac{e^{j\cdot 7Wt} + e^{-j\cdot 7Wt}}{2}\right]$$



$$z_1(jw) = \frac{X(j(w-5W)) + X(j(w+5W)) + Y(j(w-7W)) + Y(j(w+7W))}{2}$$

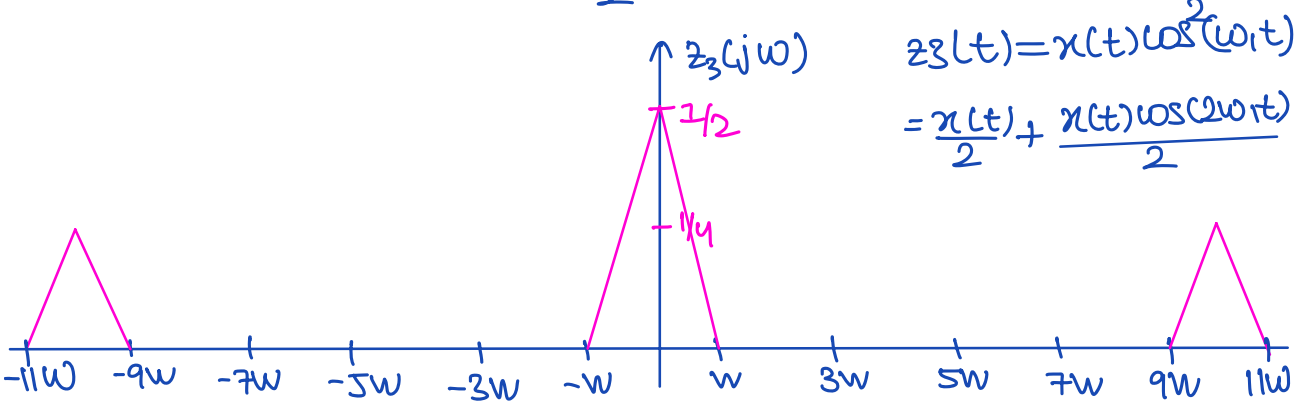


$z_2(jw)$ is $z_1(jw)$ limited from 4W to 6W



$$z_3(t) = z_2(t) \cdot \cos \omega_1 t = z_2(t) \frac{e^{j5\omega t} + e^{-j5\omega t}}{2}$$

$$\Rightarrow z_3(j\omega) = \frac{z_2(j(\omega - 5\omega)) + z_2(j(\omega + 5\omega))}{2}$$

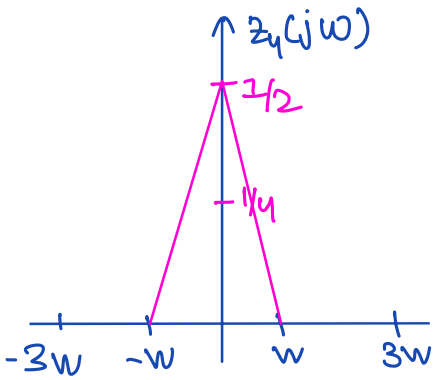


$$z_3(t) = x(t)\cos^2(\omega_1 t)$$

$$= \frac{x(t)}{2} + \frac{x(t)\cos(2\omega_1 t)}{2}$$

$x_4(t)$ is $z_3(t)$ but with max frequency w

$\Rightarrow$



$$z_4(t) = \frac{x(t)}{2}$$